

Semantic Coherence Verification for Distributed AI Inference: Validation of the Omega Metric Using Multilingual Sentence Embeddings

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Preprint · May 15, 2026

Abstract

We present the experimental validation of Omega (Ω), a semantic coherence metric designed to verify fragment consistency in distributed AI inference networks. In such networks, multiple volunteer nodes process fragments of a query independently, and a reintegrator must determine whether the returned fragments are semantically consistent before assembling a final response. We demonstrate that the multilingual sentence embedding model *paraphrase-multilingual-MiniLM-L12-v2* discriminates coherent from divergent fragment pairs with AUC-ROC of **0.9539** on a dataset of 1,000 real Wikipedia fragment pairs in Spanish, significantly exceeding the 0.85 threshold required for system viability. We further show that dataset quality is the critical factor: the same model achieves only 0.55 AUC on synthetic data, rising to 0.91 on manually curated pairs and 0.95 on Wikipedia-sourced pairs. These results validate the core hypothesis of Phase 0 and establish a reproducible baseline for distributed semantic coherence verification.

Keywords: distributed inference, semantic coherence, sentence embeddings, AUC-ROC validation, volunteer computing, decentralized AI

1. Introduction

Contemporary large language models (LLMs) require significant computational resources concentrated in centralized data centers controlled by a small number of organizations. This concentration limits access for researchers, institutions, and communities in regions without the capital to procure such infrastructure.

Distributed inference systems, such as Petals [1] and Bittensor [2], have demonstrated that volunteer computing networks can collectively run models too large for any single consumer machine. However, these systems lack a mechanism to verify semantic coherence between fragments processed by different nodes — a critical requirement when nodes may be unreliable, misconfigured, or malicious.

We propose the Omega (Ω) metric: a fast, computable measure of semantic coherence between two text fragments. In the proposed distributed architecture, a reintegrator node uses Ω to accept or

reject fragment pairs before assembling them into a final response. If Ω falls below a threshold (0.65 in our formulation), the fragment is rejected and rerouted through the ACO scheduler to an alternative node.

This paper reports the Phase 0 experiment: empirical validation that Ω can reliably discriminate coherent from divergent fragment pairs, using real-world Wikipedia data in Spanish.

2. The Omega Metric

The Ω metric combines multiple semantic signals into a single coherence score. In its initial formulation:

$$\Omega(A, B) = \alpha \times \cos(A, B) + \beta \times (1 - \text{KL}(A, B)) + \gamma \times \text{NLI}(A, B)$$

where $\cos(A,B)$ is the cosine similarity between semantic embeddings of fragments A and B; $\text{KL}(A,B)$ is the KL divergence between token probability distributions; $\text{NLI}(A,B)$ is a Natural Language Inference score (1.0 for entailment, 0.5 for neutral, 0.0 for contradiction); and α , β , γ are weights summing to 1.0.

For the Phase 0 experiment, we focus on the cosine similarity signal using multilingual sentence embeddings, which proved sufficient to exceed the 0.85 AUC threshold on real-world data. Full Ω with NLI is proposed for Phase 1 validation.

3. Experimental Setup

3.1 Dataset Construction

We constructed three datasets of fragment pairs to systematically evaluate the effect of data quality on metric performance:

Dataset	Source	Pairs	Balance
Synthetic	Rule-based generation	995	400/595
Manual	Human-authored pairs	45	25/20
Wikipedia	wikimedia/wikipedia 20231101.es	1,000	500/500

Table 1. Dataset summary. Balance column shows coherent/divergent pair counts.

For the Wikipedia dataset, coherent pairs consist of consecutive sentences from the same article (*etiqueta*=1), while divergent pairs consist of sentences from different articles (*etiqueta*=0). This construction ensures ground-truth labels without manual annotation.

3.2 Model and Evaluation

We evaluated the multilingual model *paraphrase-multilingual-MiniLM-L12-v2* from the sentence-transformers library [3]. This model produces 384-dimensional embeddings trained on paraphrase data across 50+ languages including Spanish. Cosine similarity between embeddings

serves as the Ω proxy in this experiment.

Primary metric: AUC-ROC, measuring the model's ability to rank coherent pairs above divergent pairs. We also report mean cosine similarity (μ) per class and class separation ($\mu_{\text{coherent}} - \mu_{\text{divergent}}$). The viability threshold is $\text{AUC} \geq 0.85$, below which the metric is insufficient for reliable fragment verification in production.

All experiments were run on Google Colab with a Tesla T4 GPU (16GB VRAM). Total GPU time: approximately 5 minutes for the Wikipedia experiment.

4. Results

Table 2 summarizes AUC-ROC across all three dataset conditions:

Dataset	AUC-ROC	μ coherent	μ divergent	Separation	Status
Synthetic	0.5592	0.613	0.573	0.040	FAIL
Manual (45 pairs)	0.9060	0.842	0.365	0.477	PASS
Wikipedia (1,000)	0.9539	0.433	0.064	0.369	PASS

Table 2. AUC-ROC results across dataset conditions. Viability threshold: $\text{AUC} \geq 0.85$.

Wikipedia dataset: AUC-ROC = 0.9539 | Separation = 0.369 | APROBADO

The Wikipedia experiment yields AUC-ROC of 0.9539, substantially above the 0.85 threshold. The mean cosine similarity for coherent pairs (0.433) is 6.8x higher than for divergent pairs (0.064), indicating strong and reliable discrimination.

The synthetic dataset achieves only 0.5592 AUC, barely above random. Analysis reveals that synthetic coherent pairs (adjacent sentences in a short corpus) are not semantically similar enough to be distinguished from synthetic divergent pairs (sentences from different domains), because the corpus vocabulary is homogeneous. This demonstrates that dataset construction methodology is the primary determinant of experimental validity.

The manual dataset (45 pairs) achieves 0.906 AUC with a separation of 0.477, confirming that the embedding model performs well when pairs are carefully constructed with clear semantic contrast. The Wikipedia dataset scales this approach to 1,000 pairs with naturally occurring text, achieving 0.9539 AUC.

5. Discussion

5.1 Implications for Distributed AI

The validation of Ω enables the construction of a distributed inference network where fragment coherence can be verified automatically and efficiently. The sentence-transformers model requires approximately 2ms per pair on GPU, well within the latency budget of the proposed QoS architecture (target: <200ms end-to-end for Phase 2).

The strong discrimination achieved (AUC 0.9539) suggests that even a malicious node returning plausible-sounding but semantically divergent content would be detected with high probability before integration into the final response.

5.2 Limitations

The current validation uses sentence-level pairs from encyclopedia text. Production fragments will vary in length, domain, and linguistic complexity. The metric must be validated on domain-specific corpora (medical, legal, code) before deployment in those contexts. Additionally, adversarial attacks — where a malicious node produces text that is semantically similar but factually incorrect — are not captured by this evaluation and require separate study.

The experiment uses cosine similarity as the sole Ω signal. The full formulation incorporates KL divergence and NLI, which may improve discrimination of the hardest cases (same-domain, opposite-argument pairs, subgroup C in our taxonomy). This is the primary objective of Phase 1 validation.

5.3 Dataset Quality as the Critical Factor

The 40-point AUC gap between synthetic (0.56) and Wikipedia (0.95) data is the most significant finding of this experiment. It implies that researchers validating semantic coherence metrics must use naturally occurring text, not rule-generated synthetic pairs. We provide the Wikipedia dataset and generation code as open artifacts to enable reproducible evaluation.

6. Related Work

Distributed LLM inference has been explored by Borzunov et al. [1] (Petals), which distributes transformer layers across volunteer nodes but does not address semantic verification between nodes. Bittensor [2] introduces economic incentives for distributed inference but uses benchmark accuracy rather than pairwise semantic coherence for quality assessment. PARALLAX [7] proposes a two-phase scheduler for volunteer GPU pools optimizing latency and throughput, but similarly does not include a semantic coherence verification mechanism between nodes.

The most closely related concurrent work is WISV [8] (April 2025), which proposes wireless-informed semantic verification for distributed speculative decoding. WISV integrates channel state information (CSI) with hidden representations to accept or reject speculative tokens in device-edge inference. Our work differs in three key aspects: (1) we target volunteer-node networks over general internet rather than device-edge wireless links; (2) we use pairwise sentence embeddings rather than internal hidden states augmented with CSI; and (3) we validate using AUC-ROC on a labeled dataset rather than throughput metrics.

Semantic textual similarity using sentence embeddings is well-established [3,4]. Our contribution is the application of this technology to the specific problem of fragment coherence verification in volunteer-node distributed inference, and the empirical demonstration that it achieves the AUC threshold required for system viability. We further identify dataset construction quality as the primary determinant of metric validity, a finding not reported in prior work.

The Ω formulation draws from consensus mechanisms in distributed systems (Raft [5], PBFT [6]) and applies analogous voting logic to semantic coherence rather than exact value agreement. Lee et al. [9] apply pairwise cosine similarity between model update feature vectors to detect Byzantine nodes in federated learning — a structurally similar idea applied to model training rather than inference.

7. Conclusion

We have validated that semantic coherence between distributed AI inference fragments can be reliably measured using multilingual sentence embeddings. On 1,000 real Wikipedia fragment pairs in Spanish, the AUC-ROC of 0.9539 substantially exceeds the 0.85 viability threshold, with a class separation of 0.369.

This result constitutes Phase 0 approval for the distributed AI inference network project. The next phase will implement a 3-node prototype with Tailscale mesh networking and validate the full Ω metric incorporating KL divergence and NLI signals on domain-specific corpora.

All code, datasets, and results are available at: [github.com/\[repository\]](https://github.com/[repository]) (to be published)

Acknowledgments

The author thanks the open-source communities behind sentence-transformers, Hugging Face, and the Wikimedia Foundation for providing the tools and datasets that made this experiment possible. The conceptual architecture of the distributed inference system was developed iteratively with the assistance of AI language models, which contributed to the design of the QoS framework, ACO scheduler, and the identification of analogous technologies from aerospace, finance, and telecommunications domains.

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